

2020FAPS110TestSAQ

Wednesday, October 28, 2020 7:33 PM

UNIVERSITY OF TORONTO

FACULTY OF APPLIED SCIENCE AND ENGINEERING

Short Answer Assessment

Due: October 29th 2020 by 2359 Eastern Daylight Time (Toronto)

First Year

APS110 ENGINEERING CHEMISTRY AND MATERIALS SCIENCE

Exam Type: X (Open Book; Internet, if not communicated/posted during test; no communication with others)

Examiners: M Kortschot, C Jia, S Ramsay

All numerical responses must be expressed in the most appropriate units, including appropriate prefixes (ex. GPa rather than 10^9 Pa), and be written with appropriate significant figures.

Note: as this is an official assessment, late submissions will not be accepted. This means that if you do not submit your responses successfully to Crowdmark by the deadline you will not be able to submit and you will need to submit an official petition. Therefore, it is highly advisable that you make plans to complete and upload your final PDF at least 2-3 hours early.

Please generate your answers as a single PDF document and upload to Crowdmark.

Question	Points	Score
1	10	
2	10	
3	15	
4	10	
Total:	45	

- 8 1. (10 points) Based on the popularity of another similar promotion, a restaurant is running an advertising promotion to increase orders through its delivery operation during the pandemic. The restaurant has filled its unused dining room with footballs (soccer balls) and softballs, having diameters of 220 mm and 97 mm, respectively. To the person who can estimate most closely the number of each type of ball in the room, the restaurant will award a lifetime supply of sushi pizza. The balls are packed into the room in such a manner that each softball is touching exactly six footballs (soccer balls), the footballs (soccer balls) are not touching one another, and each type of ball is homogeneously distributed throughout the room. If the room has dimensions of 11 m $\times$ 3.4 m $\times$ 8.1 m, estimate the number of each type of ball in the room.

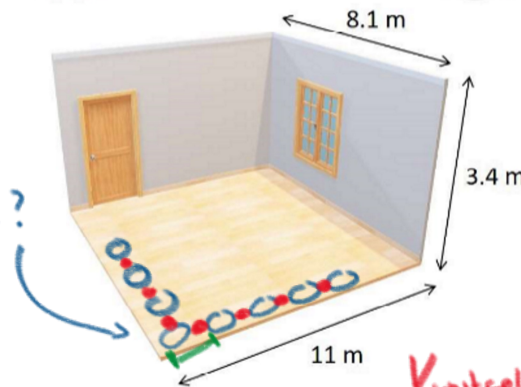




$$a = 2R_A + 2R_C$$

approach 1

how many balls along each wall?



either approach okay.

Setting up the problem (2)

$$V_{\text{unit cell}} \approx \#/\text{wall length} \quad (4)$$

Final answer (4)

$$\frac{11\text{m}}{(0.097 + 0.22)} = 34.7 \text{ pairs} = 69.401 \text{ balls}$$

$$3.4\text{m} \rightarrow 10.7 \text{ pairs} = 21.451 \text{ balls}$$

$$8.1\text{m} \rightarrow 25.52 \text{ pairs} = 51.04 \text{ balls}$$

$$\text{so, one horizontal layer has } (69.401)(51.04) = 3.5466(10^3) \text{ balls}$$

$$\text{and } 21.451 \text{ layers, so } 7.6079(10^4) \text{ balls, half footballs, half softballs}$$

$$\rightarrow \boxed{38040 \text{ of each ball}}$$

Approach 2

$$\text{Volume} \rightarrow \frac{V_{\text{Room}}}{V_{\text{unit cell}}} \cdot \#_{\text{Balls per cell}} = \# \text{ Balls in room.}$$

$$V_{\text{unit cell}} = a^3 = (220(10^{-3}) + 97(10^{-3}))^3 = 3.1855(10^{-2})$$

$$V_{\text{Room}} = (8.1 \cdot 3.4 \cdot 11) = 302.94 \text{ m}^3$$

$$\#_{\text{Balls per cell}} = 4 \text{ of each.}$$

$$\# \text{ Balls in room} = \frac{302.94}{3.1855(10^{-2})} \cdot 4 = \boxed{38040 \text{ of each Ball}}$$

2. (10 points) Sketch a series of at least 5 2D sketches showing how dislocation movement is the mechanism for plastic deformation. Be sure to indicate clearly any bonds that have broken, or formed.

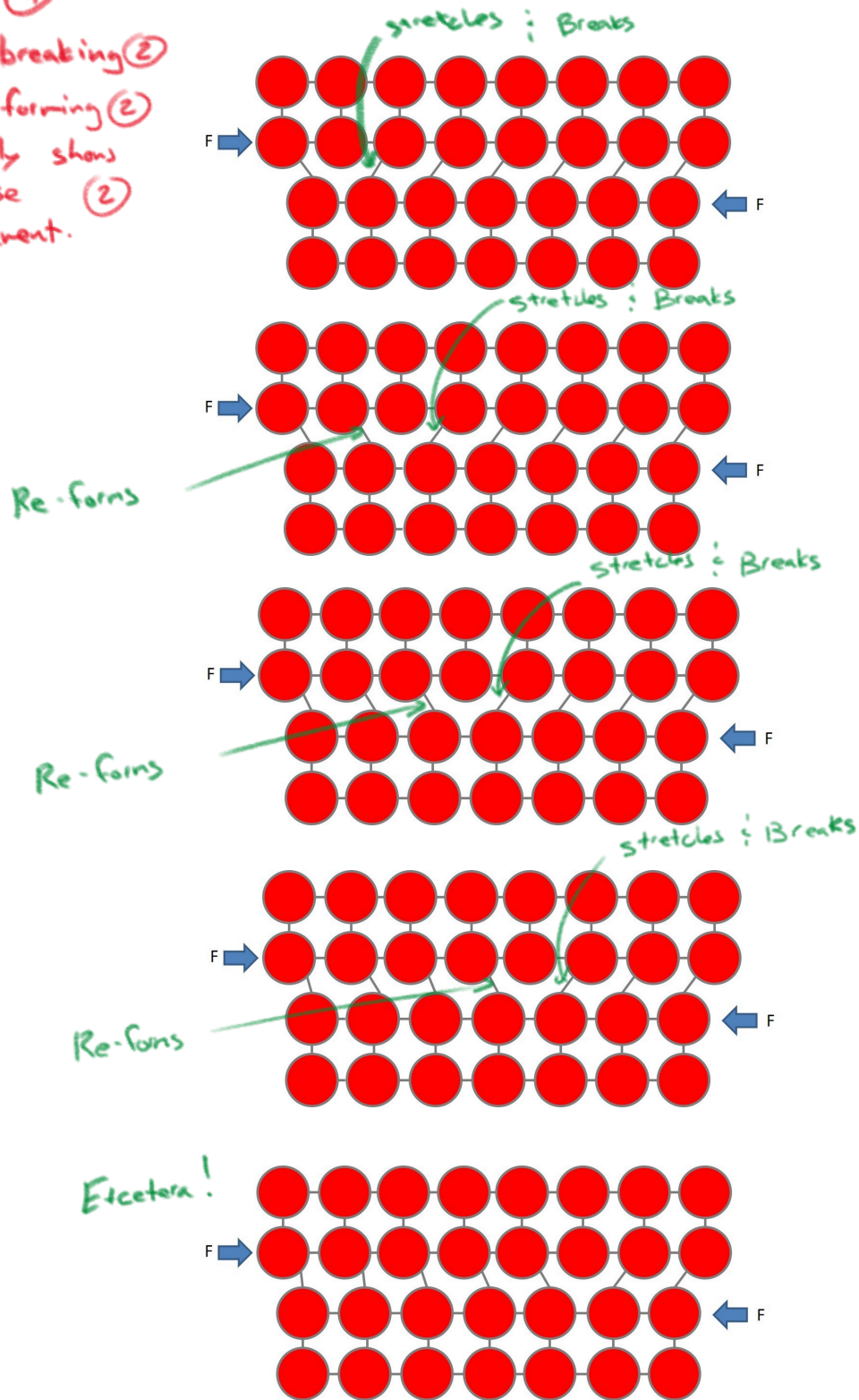
2. (10 points) Sketch a series of at least 5 2D sketches showing how dislocation movement is the mechanism for plastic deformation. Be sure to indicate clearly any bonds that have broken, or formed.

- Sketches (4)

- Bonds breaking (2)

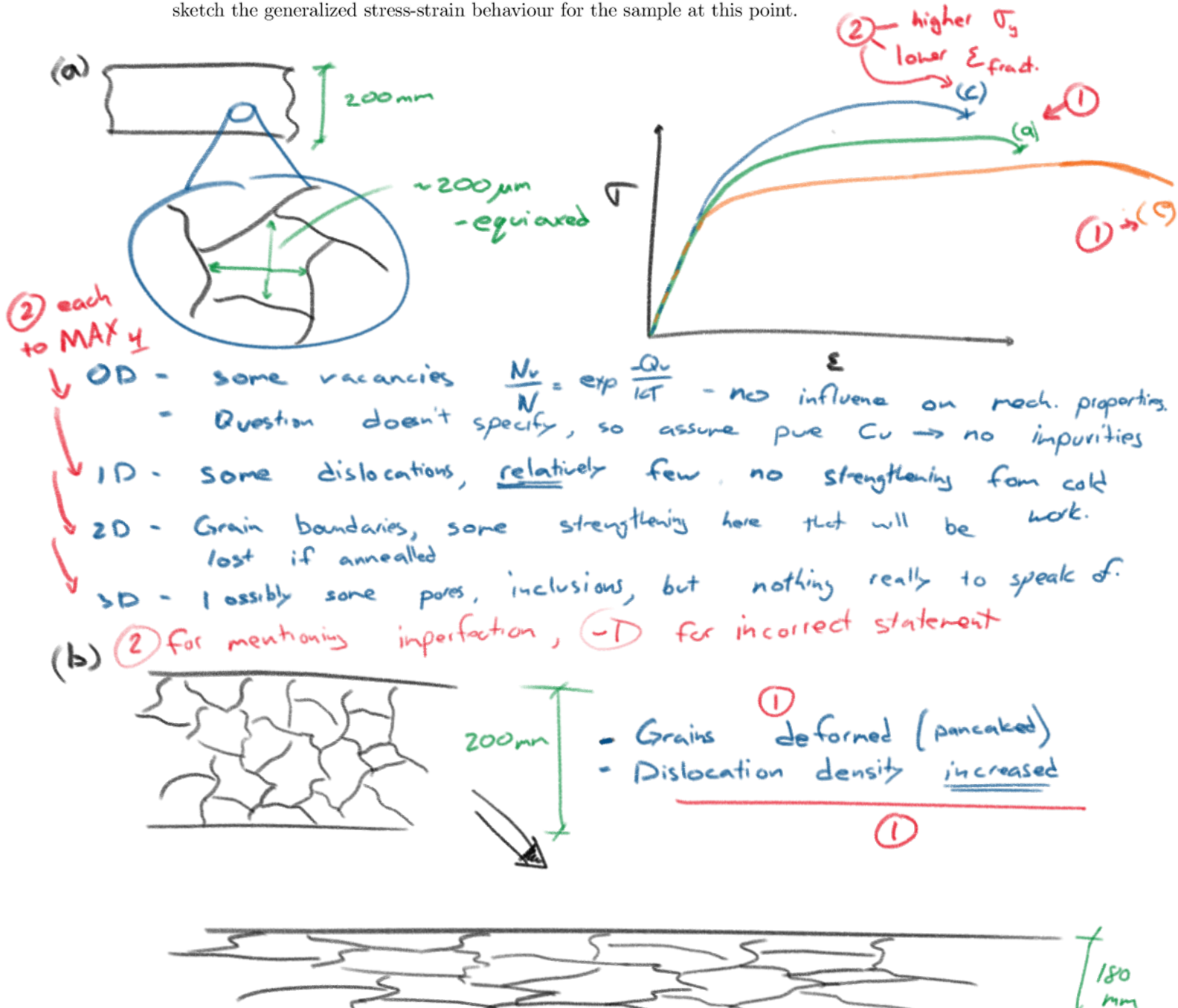
- " forming (2)

- correctly shows
Stepwise movement. (2)



3. This question has several parts. Take a moment to read the entire question before beginning to answer any of the parts of this question. Answer this question within the context of topics covered in this course so far. A sample of copper is melted, poured into a mould and allowed to solidify, resulting in a 200 mm thick bar of copper with grains that are roughly 200 μm in diameter.

- (5 points) Describe the microstructure (features around the atomic scale, changes to crystalline imperfections, etc.) that you would expect within this copper sample. Sketch the generalized stress-strain behaviour for the sample at this point.
- (2 points) This copper sample is then passed through large rollers to reduce its thickness to 180 mm. Describe the changes that will occur to the microstructure of this copper sample.
- (5 points) Explain how the microstructural changes that you identified in part (b) will lead to changes in the mechanical properties of this copper. On the same axes as your answer in part (a), sketch the generalized stress-strain behaviour for the sample at this point.
- (1 point) This copper sample is then heated to 800°C for 10 hours. Describe the changes that will occur to the microstructure of this copper sample.
- (2 points) Explain how the microstructural changes that you identified in part (d) will lead to changes in the mechanical properties of this copper. On the same axes as your answer in part (a), sketch the generalized stress-strain behaviour for the sample at this point.





- (c) - increased dislocation density will increase yield strength. ①
 - Young's Modulus unchanged ①
 - decreased ductility. ①
- (d) - decreased dislocation density ①, either
 - increased grain size
- (e) - strength will decrease
 - Young's modulus unchanged } ①, any 2.
 - increased ductility

4. (10 points) Data obtained from some random internet sources have been tabulated in Table 4. Using the data in the table, calculate the missing data for each element. Show your work.

each
 ① appropriate approach.
 ① correct answer.

Element/Compound	Radius (pm)	Crystal Structure	Molar Mass (g/mol)	Density (g/cm ³)
Copper		FCC	63.546	8.96
Cobalt	200	HCP	58.933	
Polonium	168	Simple Cubic		9.196
Scandium	162	HCP	44.955	
Silver Chloride	$R_{Ag} = 144$ $R_{Cl} = :$	Rock Salt	$A_{Ag} = 107.87$	5.56

Cl: 35.453

This is the end of the written portion of the assessment.

Cu FCC $\rho = \frac{n \cdot A}{a^3 \cdot N_A}$ | $a = 2\sqrt{2}R$ | $\rho = \frac{n \cdot A}{(2\sqrt{2}R)^3 \cdot N_A}$

$$2\sqrt{2}R = \left(\frac{n \cdot A}{\rho \cdot N_A} \right)^{1/3} \Rightarrow R = \left(\frac{n \cdot A}{\rho \cdot N_A} \right)^{1/3} \cdot \frac{1}{2\sqrt{2}}$$

$$R = \left(\frac{4 \cdot 63.546}{8.96(10^6) \cdot 6.022(10^{23})} \right)^{1/3} \cdot \frac{1}{2\sqrt{2}} = 1.2769(10^{-10})m$$

127 pm

Co HCP (take FCC shortcut)

$$\rho = \frac{n \cdot A}{V_c \cdot N_A} \quad \left| \begin{array}{l} n=4 \\ A=58.933 \end{array} \right. \quad V_c = (2\sqrt{2}R)^3 \quad \left| \quad R = 200(10^{-12})m \right.$$

$$= \frac{4 \cdot 58.933}{(2\sqrt{2} \cdot 200(10^{-12}))^3 \cdot 6.022(10^{23})} = 2.1625(10^6) \frac{g}{m^3} \cdot \left(\frac{1m}{100cm} \right)^3$$

$$= \frac{4 \cdot 58.933}{(2\sqrt{2} \cdot 200(10^{-12}))^3 \cdot 6.022(10^{23})} = 2.1625(10^6) \frac{\text{g}}{\text{m}^3} \cdot \left(\frac{1\text{m}}{100\text{cm}}\right)$$

$$\boxed{= 2.2 \frac{\text{g}}{\text{cm}^3}}$$



$$n = \left(\frac{1}{8} \cdot 8 + \frac{1}{2} \cdot 2\right) \cdot 2 + 3$$

$$= 6$$

assume $\rightarrow$

$$V_c = \frac{3\sqrt{3}}{2} a^2 \cdot h \quad \left| \begin{array}{l} h = 1.633a \\ a = 2R \end{array} \right.$$

$$\rho = \frac{6 \cdot 58.933}{\frac{3\sqrt{3}}{2} \cdot 1.633 (2 \cdot 200(10^{-12}))^3 \cdot 6.022(10^{23})} = 2.1625(10^6)$$

$$= 2.2 \frac{\text{g}}{\text{cm}^3}$$

Note actual density is $\sim 8.9 \frac{\text{g}}{\text{cm}^3}$ which is what you get with $R = 125 \text{ pm}$, which is a closer radius. But the question did specify "... random ..." internet sources!

Po  $n = 1 \quad \left| \begin{array}{l} R = 168(10^{-12}) \\ a = 2R \end{array} \right. \quad \rho = \frac{n \cdot A}{V_c \cdot N_A} \Rightarrow A = \frac{\rho \cdot V_c \cdot N_A}{n} \quad \left| \begin{array}{l} \rho = 9.196 \frac{\text{g}}{\text{cm}^3} \\ = 9.196(10^6) \frac{\text{g}}{\text{m}^3} \end{array} \right.$

$$V_c = a^3 \quad \left| \quad A = \frac{9.196(10^6) \cdot (2 \cdot 168(10^{-12}))^3 \cdot 6.022(10^{23})}{1} \right.$$

$$= 210.07 \frac{\text{g}}{\text{mol}} = \boxed{210 \frac{\text{g}}{\text{mol}}}$$

Sc use FCC shortcut

$$\rho = \frac{nA}{V_c \cdot N_A} = \frac{nA}{(2\sqrt{2}R)^3 \cdot N_A} = \frac{4 \cdot 44.955}{(2\sqrt{2} \cdot 162(10^{-12}))^3 \cdot 6.022(10^{23})} = 3.104(10^6)$$

$$\frac{\text{g}}{\text{m}^3}$$

$$= \boxed{3.1 \frac{\text{g}}{\text{cm}^3}}$$

AgCl $\rho = \frac{n_A A_A + n_C A_C}{V_c \cdot N_A} \quad \left| \quad V_c = a^3 \quad \left| \quad a = 2R_A + 2R_C \right. \right.$

$$\rho = \frac{n_A A_A + n_C A_C}{(2R_A + 2R_C)^3 \cdot N_A} \Rightarrow (2R_A + 2R_C) = \left(\frac{n_A A_A + n_C A_C}{\rho \cdot N_A} \right)^{1/3}$$

$$\rho = \frac{(2R_A + 2R_C)^3 \cdot N_A}{\left(\frac{n_A A_A + n_C A_C}{\rho \cdot N_A} \right)^{1/3} - 2R_C} \cdot \frac{1}{2} \quad \left(\rho \cdot N_A \right)$$

$$R_A = \left[\left(\frac{n_A A_A + n_C A_C}{\rho \cdot N_A} \right)^{1/3} - 2R_C \right] \cdot \frac{1}{2} \quad \left| \begin{array}{l} n_A = 4 \\ n_C = 4 \\ \rho = 5.56(10^6) \frac{\text{g}}{\text{cm}^3} \end{array} \right. \quad \begin{array}{l} R_C = 144(10^{-12}) \text{m} \\ A_A = 35.453 \\ A_C = 107.87 \end{array}$$

$$R_A = \left[\left(\frac{4 \cdot 35.453 + 4 \cdot 107.87}{5.56(10^6) \cdot 6.022(10^{23})} \right)^{1/3} - 2 \cdot 144(10^{-12}) \right] \cdot \frac{1}{2}$$

$$= 1.3365(10^{-10}) \text{m} = 133 \text{ pm}$$

aside

Note: 144 pm is atomic radius, ionic radius of cation is smaller, closer to 129 pm, and anion radius should be larger, closer to 167 pm